

A Multimodal Approach to Predicting Chronic Kidney Disease Progression Using Kidney Ultrasound Features and Longitudinal Clinical Biomarkers.

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Abstract— Chronic Kidney Disease (CKD) is a long-term condition that gradually worsen over time and regular monitoring needs to prevent severe complications and improve patient outcomes. However, most existing prediction approaches rely mainly on individual clinical parameters and often fail to capture the combined structural and temporal changes associated with disease progression. In this study, we propose a multimodal framework that integrates ultrasound imaging features with clinical laboratory data for CKD progression prediction. Kidney structural information is extracted from ultrasound images using a U-Net, while key clinical parameters as estimated glomerular filtration rate (eGFR) and serum creatinine are obtained from lab reports using Tesseract OCR, with optional manual input to improve reliability. These features are then combined and use Long Short-Term Memory (LSTM) network, enabling the system to learn temporal patterns in patient data over time. The proposed model achieves an accuracy of approximately 78% for next-visit prediction and 74% for 6-month progression forecasting, demonstrating its capability to capture both short-term and longer-term disease trends. And segmentation obtained 97.63% accuracy. Overall, this work presents a practical and scalable approach for CKD prediction, with potential applications in clinical decision support systems to assist healthcare professionals in early intervention and patient management.

Keywords— Chronic Kidney Disease (CKD), Long Short-Term Memory (LSTM), Ultrasound(US), estimated glomerular filtration rate (eGFR), Optical Character Recognition (OCR)

I. INTRODUCTION

Chronic Kidney disease (CKD) is major worldwide health issue characterized by the gradual loss of a kidney function and often progress till advanced stages. In 2017, the global prevalence was 9.1%, with approximately 700 million cases [1]. Timely intervention is critical to early detection and accurate disease progression prediction while reducing morbidity and improving patients' health conditions,

However traditional approaches primarily rely on isolated laboratory markers such as estimated glomerular filtration rate (eGFR) and serum creatinine. These markers are informative but often fail to capture the long-term temporal and structural changes in the kidney which are associated with CKD progression.

Recent advances in Artificial Intelligence (AI) and Deep learning have demonstrated advanced capabilities in medical diagnosis and outcome prediction. Notably longitudinal modeling techniques such as Long Short-Term Memory (LSTM) networks enable the analysis of patients' longitudinal data. The LSTM network deals with sequential dependencies and learns temporal trends in the development of the disease [9]. And medical imaging, specifically ultrasound (US) provides essential structural information in kidneys such as kidney length which is an indicator of renal health. Despite these advancements, existing studies focus on either clinical data or imaging data, lacking unified framework that can improve prediction accuracy. To address this limitation this study proposes a multimodal framework which is a combination of ultrasound image analysis and clinical data extraction to predict the probability of CKD stage progression. Specifically, kidney structural features extraction such as kidney length by using U net-based segmentation model, and obtain clinical parameters such as eGFR, serum creatinine, age, gender through Optical Character recognition (OCR) from medical laboratory reports. Then utilize an LSTM network to capture both spatial and temporal patterns in the patient's data. Furthermore, it integrates the automatically derived ultrasound-based kidney length with laboratory report values through data fusion. This study aims to predict the probability of progression to the next CKD stage for both the subsequent clinical visits, as well as six-month future time window.

II. RELATED WORK

Chronic Kidney Disease (CKD) has been widely studied due to its increasing global impact, and many researchers have explored different ways to improve early detection and predict disease progression.

Early researches, mostly relied on clinical biomarkers like eGFR to predict current stage in CKD and monitor CKD progression. Common indicators such as serum creatinine, estimated glomerular filtration rate (eGFR), and albuminuria were widely used to evaluate kidney function [2]. Over time, researchers began exploring additional biomarkers like NT-proBNP and cardiac troponins, which showed strong relationships with disease progression and patient outcomes [3], [4]. These studies helped improve early diagnosis and provided better ways to assess risk, especially in complex or unclear cases [5], [6].

As data-driven approaches became more popular, machine learning techniques started playing a major role in CKD prediction. Algorithms such as Decision Trees, Support Vector Machines (SVM), and Random Forests were applied to clinical datasets and achieved promising results in identifying current CKD stages [7]. Researchers also found that selecting the most relevant features from the data significantly improved model performance [8]. In addition, ensemble methods were introduced to make predictions more stable and reliable across different datasets.

Deep learning models make big impact on recent studies as deep learning can automatically learn complex patterns from data. While some approaches integrate Convolutional Neural Networks (CNN) with Long Short-Term Memory (LSTM) networks to jointly learn image and sequential patterns [9]. In medical imaging, architectures like U-Net have been successfully used to segment kidney regions from ultrasound images, making it possible to extract important structural features such as kidney size and shape [10]. Other methods focus on multimodal approaches as extracting structural features from medical images using segmentation models, and also then applying temporal models like LSTM for disease progression prediction.

Another important direction in CKD research is temporal modeling, since the disease progresses over time. LSTM-based models have been widely used to analyze sequential patient records and predict how the disease evolves over time [11]. Some studies also addressed real-world challenges like missing or irregular medical data by proposing methods that can handle incomplete Electronic Health Records (EHRs) [12]. Additionally, techniques such as Natural Language Processing (NLP) have been used to extract useful information from clinical notes, further improving prediction accuracy [13].

Recent research has shown that deep learning models are highly effective for predicting Chronic Kidney Disease (CKD). A study by Akter *et al.* [14]. compared several models such as ANN, LSTM, GRU, and RNN using clinical

data, and reported very high accuracy, with ANN achieving up to 99%. The study also highlighted the importance of feature selection and risk factor analysis in improving prediction performance. However, the approach mainly relied on structured clinical data and did not consider imaging features, indicating the need for more comprehensive methods that combine different types of data. Explainable AI methods focus on making model decisions more transparent and trustworthy for clinical use [15].

There are also studies that consider region-specific factors, especially in countries like Sri Lanka, where CKD has unique environmental and demographic influences [16]. Furthermore, some research has focused on developing mobile health applications to support patients in managing their condition, showing how these technologies can be applied in real-world healthcare settings [17].

Despite all these advancements, several limitations still exist. Many studies focus only on either clinical data or imaging data, rather than combining both. Also, predicting future CKD progression over time using a multimodal approach remains a challenge, especially when dealing with complex and incomplete patient data.

III. METHODOLOGY

The proposed system is designed to predict the probability of chronic kidney disease (CKD) progression. This system integrates ultrasound-based structural feature extraction using U-Net segmentation with OCR-based clinical data extraction. The fused features are modeled using an Long Short-Term Memory (LSTM) network to capture temporal disease progression patterns. AND the overall pipeline consists of ultrasound image processing, laboratory data extraction and feature fusion, and prediction using LSTM network. This multimodal framework captures both structural and clinical biomarkers to improve prediction accuracy.

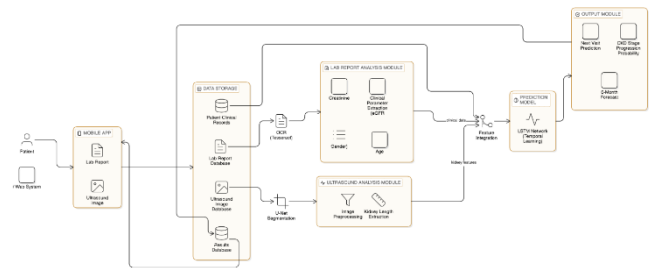


Fig. 1. Architectural diagram,

Fig. 1, illustrate overall architecture of the proposed system. Representing a multimodal CKD progression prediction system where patient data like ultrasound and lab reports collected by mobile application. The ultrasound images are processed using a U-Net model to extract the kidney length, while lab reports key clinical parameters like eGFR, creatinine, age, gender are extracted through Tesseract OCR. These features then combined and fed into LSTM model. LSTM models capture temporal patterns in data to predict the CKD Stage progression probability in next visit outcome and 6-month forecast.

A. Data Collection

The data set used in this study was obtained from reliable sources to support multimodal CKD progression prediction. Clinical and temporal patient data were taken from the publicly available MIMIC-IV (Medical Information Mart for Intensive Care) database [18], which contains de-identified electronic health records of patients admitted to intensive care units. The dataset provides longitudinal information such as estimated glomerular filtration rate (eGFR), serum creatinine levels, demographic details, and time-stamped clinical records. These data are particularly useful for understanding how a patient's condition changes over time, making them well suited for training the LSTM model.

For ultrasound image analysis, we used kidney ultrasound images along with their corresponding segmentation masks from the publicly available Kaggle dataset, "US Simulation & Segmentation Dataset" [19]. This dataset includes both synthetic and real ultrasound images, which makes it suitable for training segmentation models to capture important structural features of the kidney in a more robust way. Additionally, laboratory reports were processed using Optical Character Recognition (OCR) to extract key clinical values such as estimated glomerular filtration rate (eGFR) and serum creatinine levels. In cases where the extracted information was incomplete, unclear, or affected by OCR errors, manual checking and correction were carried out to ensure the final dataset remained accurate and reliable.

Although the datasets originate from different sources they utilized in a complementary manner within the proposed framework. The MIMIC-IV dataset is exclusively used to train the LSTM based model. while ultrasound derived kidney length and OCR extracted clinical features used as auxiliary inputs. These features integrated at the feature level to enable comprehensive CKD progression probability prediction.

B. Data Preprocessing

Ultrasound images are processed to improve quality and ensure consistency before the extraction of the feature. Resizing images to a fixed resolution (256*256), noise reduced using filtering techniques and used grayscale conversion, intensity normalization to improve segmentation accuracy and reduce variability across different ultrasound machines.

Lab reports often in unstructured image format therefore used Optical Character Recognition (OCR) technology to transformed picture format to digital format. And also extracted important textual data from the images including eGFR, serum creatinine. And missing values handled using imputation methods. All numerical features are normalized to a standard to facilitate model training.

C. Feature extraction

Ultrasound images are processed using a U net segmentation model to isolate the kidney region. From the segmented mask, the kidney boundary is estimated by computing the maximum distance between edge points. This pixel value then converts to real world units using calibration factor to get the kidney length.

$$L_{\text{kidney}} = D_{\text{pixel}} * R$$

D_{pixel} represent the maximum ferret diameter in pixels and R is the Pixel to centimeter converted ratio. Resulting kidney length features are used as a key feature for CKD progression prediction.

Clinical data is extracted from lab reports using Tesseract OCR. The extracted parameters are eGFR and Serum creatinine. The age and gender is a static feature. these features represent functional kidney health indicators.

D. CKD progression prediction model

The proposed system uses a Long Short-Term Memory (LSTM) model capture temporal dependencies in patient data. LSTM model process sequential clinic data to learn the disease progression pattern over time. The LSTM captures long term dependencies in patient history through internal gates. Enabling effective learning of disease evolution over time.

Final prediction is computed using a SoftMax layer as follows:

$$P(y) = \text{Softmax}(W \cdot h_t + b)$$

h_t represents the hidden state of the LSTM at time stamp t , W is the weight matrix, and b is the bias term. The output represents the probability of distribution over CKD stages indicated the likelihood of disease progression at future clinical visits.

E. Model training and performance evaluation

The kidney segmentation is performed using a U net model on annotated ultrasound image to extract kidney regions for structural analysis. In U net for loss function use Dice loss/ Binary Cross entropy and Adam optimizer with Relu activation. segmentation performance is evaluated using Dice Coefficient.

The temporal prediction model is implemented using a LSTM network to capture sequential dependencies in health data. In LSTM model, Adam optimizer, and for loss function use Categorical cross entropy. And model is evaluated using Accuracy, Precision, Recall, F1-Score and Validation loss.

IV. RESULTS AND DISCUSSION

A. Kidney ultrasound segmentation

The U-Net model was trained using 633 paired ultrasound images and corresponding segmentation masks to automate measurements of kidney structural features. During training, the model converged quickly and achieved high pixel-level accuracy but shows sensitivity to ultrasound noise.

Table I: Quantitative Metrics for U-Net Segmentation

Metric	Training value	Validation value
Accuracy	97.63%	97.83%
Binary cross entropy(Loss)	0.0740	0.0628
Dice Score	0.2303	-

The model achieved 97.67% training accuracy. And the Dice score of 0.2303 indicates that model correctly identify the

presence of the kidney , but the boundary detection remains challenging due to low contrast and noisy nature of the B-mode ultrasound images. Likely due to shadowing the artifacts and blurred edges.

B. Multimodal CKD Progression forecasting

The LSTM network was trained to predict CKD stage progression in next visit and 6 month forecast. Using multimodal integration of laboratory data and ultrasound data(Lab+US)

Table II: . LSTM Training Analysis

Prediction Task	Metric	Lab + US Mode
Next Visit	Validation Accuracy	78.35%
	Validation Loss	0.5383
6-Month	Validation Accuracy	73.65%
	Validation Loss	0.6772

Table 2. LSTM Training Analysis: The table demonstrates a multi-task learning trajectory. The Next-Visit Accuracy 78.35% outperforms the 6-Month Accuracy 73.65%.

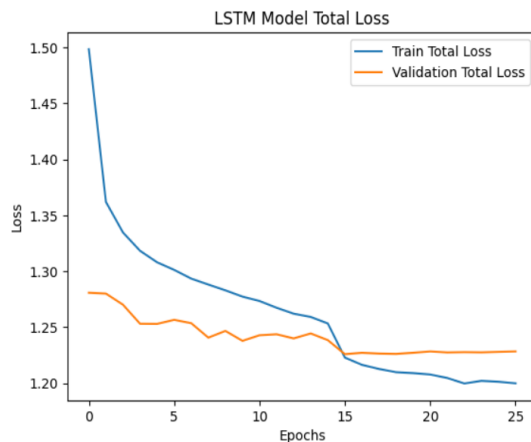


Fig 2. LSTM model Loss

Fig2. Indicates that the training and validation loss curves show a consistent decreasing trend, indicating effective learning and model convergence. The training loss curve indicates steady decreasing , while the validation loss follows a similar pattern with minor fluctuations. Both curves stabilize after approximately 15 epochs, suggesting convergence. The small gap between the two indicates minimal overfitting and good generalization performance.

The LSTM architecture’s ability to capture the eGFR slope is one of the primary drives of its forecasting accuracy. By processing sequence of 10 visits (MAX_SEQ_LEN=10) the

model identifies whether patient has rapid decline or stable impairment. The 6-month prediction allows for early intervention before patient health worsens.

V. CONCLUSION

This study presented a multimodal framework for predicting Chronic Kidney Disease (CKD) progression probability by integrating ultrasound imaging features like kidney length with clinical laboratory data. The proposed system combines kidney structural information, obtained through U-Net-based segmentation, with functional biomarkers extracted using Tesseract OCR, and models temporal disease patterns over time using a Long Short-Term Memory (LSTM) network.

The experimental results demonstrate that the model achieves a prediction accuracy of approximately 78% for next-visit CKD progression , 74% for 6-month forecasting and the segmentation model get accuracy approximately 97.63% .indicating its effectiveness in capturing both short-term and long-term disease dynamics. The integration of multimodal data enables the system to leverage complementary structural and clinical information, resulting in improved robustness compared to single-modality approaches.

Furthermore, the proposed framework is designed with real-world applicability in mind, incorporating automated data extraction alongside manual input support, making it suitable for deployment in clinical decision support systems. Despite certain limitations such as dataset size and OCR-related noise, the results highlight the potential of multimodal deep learning approaches in enhancing CKD prognosis.

Future work will focus on expanding the dataset, incorporating additional imaging and clinical biomarkers for the same patient over long time. And exploring advanced architecture such as attention-based models to further improve prediction accuracy. Overall, the proposed system provides a scalable and effective solution for early prediction of CKD progression, contributing toward improved patient management and clinical outcomes.

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